Environment and Resource Management

Vrije Universiteit Amsterdam - Faculty of Science - Environment and Resource Management - 2025-2026

The one-year Master program Environment and Resource Management (ERM) is designed to train the next generation of decision-makers to work on the most pressing sustainability challenges of our time. Students learn to analyze problems and design solutions for challenges including climate change, biodiversity loss, water scarcity and drought, and food security, among others. The program is highly international and is open for students from all disciplinary backgrounds, training them to apply their disciplinary knowledge in an multidisciplinary environment, often in teams. Students will also be familiarized with key disciplines for studying sustainability and learn to integrate different bodies of knowledge in a problem-oriented manner. Students can determine their own focus by choosing a specialization and further establish their own niche through the Master's thesis. The five specializations have been crafted to cover important contemporary sustainability challenges and include: Energy and Climate, Global Food Challenges, Global Water Challenges, Ecosystems Services and Biodiversity, and Global Sustainable Futures.

Info

Level	Master
Language	English
Duration	1 year
Type	Full Time
Credits	60 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Environment and Resource Management constrained choice

Courses

Course Name	Period	Credits	Code
Economics of Environmental Policy	P2	6EC	E_STR_EEP
Environmental Economics for ERM	P2	6EC	AM_468020

Environment and Resource Management specialisation Energy and Climate

Courses

Course Name	Period	Credits	Code
Sustainable Energy Systems	P1	6EC	AM_468018
Energy and Climate Governance	P3	6EC	AM_1155

Environment and Resource Management specialisation Ecosystems Services and Biodiversity

Courses

Course Name	Period	Credits	Code
Value of Ecosystem Services	P1	6EC	AM_468024
Governance of Ecosystem Services	P3	6EC	AM_468025

Environment and Resource Management specialisation Global Sustainable Futures

Programme Sections

- ERM spec. Global Sustainable Futures constrained choice

ERM spec. Global Sustainable Futures constrained choice

Courses

Course Name	Period	Credits	Code
Food Systems & Sustainability	P1	6EC	AM_1262
Sustainable Energy Systems	P1	6EC	AM_468018
Value of Ecosystem Services	P1	6EC	AM_468024
Water extremes: Risks and Solutions	P1	6EC	AM_1305
Energy and Climate Governance	P3	6EC	AM_1155
Food Governance & Sustainability	P3	6EC	AM_1263
Governance of Ecosystem Services	P3	6EC	AM_468025
Water Politics and Governance	P3	6EC	AM_1192

Environment and Resource Management specialisation Global Food Challenges

Courses

Course Name	Period	Credits	Code
Food Systems & Sustainability	P1	6EC	AM_1262
Food Governance & Sustainability	P3	6EC	AM_1263

Environment and Resource Management compulsory courses

Courses

Course Name	Period	Credits	Code
Research Project ERM	Ac. Year (September)	18EC	AM_468017
Causes and Consequences of Environmental Change	P1	6EC	AM_1049
Environmental Policy	P2	6EC	AM_468021
Methods of Environment and Resource Management	P4	6EC	AM_1135
Research Design	P4	6EC	AM_1136

Environment and Resource Management specialisation Global Water Challenges

Courses

Course Name	Period	Credits	Code
Water extremes: Risks and Solutions	P1	6EC	AM_1305
Water Politics and Governance	P3	6EC	AM_1192

Courses

Causes and Consequences of Environmental Change

Course Code	AM_1049
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ing. P.G.B. Chopin
Examiner	dr. ing. P.G.B. Chopin
Teaching Staff	dr. ir. C.J.E. Schulp, A. Steinmann, prof. dr. P.H. Pattberg, dr. ing. P.G.B. Chopin, dr. F.M. Béen, dr. E.J.M. Boere MSc, prof. dr. ir. P.H. Verburg
Teaching method(s)	Study Group, Seminar, Computer lab, Lecture

Course Objective

The goal of this course is to develop a common knowledge base about the causes and consequences of environmental changes and about how different disciplines perceive and address environmental problems. The course provides the opportunity to develop skills expected of master's graduates, notably literature search, efficient reading and synthesis, group work and writing skills and for grasping sustainability issues.

By the end of this course, students should be able to:

- Link the different components of socio-ecological systems: demands in terms of resources, biological processes, impacts and response driving the causes and consequences of environmental change
- Capture the links between society, economics and environmental and how these interactions drive the fate of a given system

- Characterize the type of relationship among these elements by using the different system concepts that will be viewed in the course: system boundaries, reinforcing and balancing loops, non-linearity, threshold effects, uncertainty
- Judge the relevance of indicators used to describe the systems in a real-life setting linked to decision-making in environmental processes analysis
- Reflect on concept around social sciences, e.g. societal or individual responsibility and how these concepts can impact the responses to environmental changes
- Demonstrate a capacity in collaborating in an interdisciplinary environment and contribute to a shared goal
- Develop skills in handling databases, presenting information in a synthetic and well framed way, writing in an efficient way

Course Content

Current human activities including agriculture, fishing, mining among others alter the core functions of global as well as local ecosystems. Anthropogenic climate change and the accelerating loss of biodiversity are pressing drivers of the planet change and are receiving increasing attention. However, many others local to global environmental impacts are also happening in parallel such as the degradation of ecosystems services, disturbance of the nitrogen cycles notable in link with the food chain, intensive land use change and various pollutants affecting natural ecosystems and human health. More sustainable development pathways are urgently needed. Identifying such pathways require an interdisciplinary and system understanding of the drivers and impacts of human activities to search for solution. This requires the involvement of numerous academic disciplines including natural and social sciences and appropriate assessment of current system functioning via system thinking and assessment of change of these system with appropriate indicators.

Causes and consequences of environmental change presents the breadth of environmental problems and negative impacts, using different discipline to examine causes, effects and possible solutions. It is a compulsory course in Environmental Resource Management master program as well as by students from other curricula from the social environmental or earth sciences, as an extra curricula course. The course serves to develop a system understanding of environmental, social and economic impacts using i) System Dynamic approaches (e.g., causal loop diagram building in groups), ii) scenario of climate change mitigation, iii) indicators-based assessment in groups and iv) debate on the necessity of economic growth through 4 key practical exercises proposed during the course. Those practical exercises are fed by multiple guest lectures targeting specifically planetary boundaries (biosphere integrity, climate change, land system change, biogeochemical cycles, freshwater use, pollution) and society and population as a driver of negative impacts and potential solutions through for instance appropriate governance.

Additional Information Teaching Methods

The course is worth 6 ECTS credits which corresponds to 168 hours of work per student. The course comprises two sets of activities. The first takes place in classes, where information is presented through lectures, guest presentations, workshops and debates. The second includes assignments where student’s ability to achieve the course’s objectives is tested and students get hands-on experience with some of the methods discussed during the course. Assessment involves these written assignments and the exam. Feedback opportunities are included in class activities, question hours and presentations are organized for the assignments.

Approximate time allocation:

- Class: about 44 hours (HC=16, CPR=8, WC=20)
- Reading and exam preparation: about 64 hours
- Assignments and presentation: about 60 hours

Method of Assessment

Type of assessment:

- Assignments (A) worth 40% of the final grade (minimum grade to pass the course: 5.5)
- Written exam (E) worth 60% of the final grade (minimum grade to pass the course: 5.5). One resit possibility in December/January.
- It is compulsory to attend the assignment-related working lectures, activities and practical exercises.

Literature

The course builds on a series of scientific articles, which will be announced in the course guide, as well as selected book chapters which include:

Harris, J.M. and Roach, B. (2014) Environmental and Natural Resource Economics: A Contemporary Approach. 3rd Edition. Chapter 9 (Lecture Population, Affluence, Technology) and Chapters 1+7 (Lecture Economic instruments). Required text for ERM Course on Environmental Economics (Period 2).

Additional Information Target Audience

The course is part of the MSc programme 'Environment and Resource Management' but is open to all MSc students.

Additional Information

The course is coordinated by Dr. Pierre Chopin, and lectured by:

- Dr. Pierre Chopin (p.g.b.chopin@vu.nl)
- Mrs. Adna Steinmann (a.steinmann@vu.nl)
- Dr. Nynke Schulp (nynke.schulp@vu.nl)
- Dr. Esther Boere (e.j.m.boere@vu.nl)
- Prof. Dr. Philipp Pattberg (philipp.pattberg@vu.nl)
- Prof. Dr. Jeroen van den Bergh
- Dr. Harry Aiking (h.aiking@vu.nl)
- Dr. Frederic Béen (f.m.been@vu.nl)

Economics of Environmental Policy

Course Code	E_STR_EEP
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	prof. dr. M.W. Hofkes
Examiner	dr. ir. J.H. Ansink
Teaching Staff	dr. M.J. Koetse, dr. ir. J.H. Ansink
Teaching method(s)	Study Group, Lecture

Course Objective

By the end of this course, you should be able to:

ACADEMIC AND RESEARCH SKILLS

- explain the main mechanisms of economic policy instruments for managing environmental problems
- explain the pros and cons of economic policy instruments under different conditions and in different contexts

BRIDGING THEORY AND PRACTICE

- apply this knowledge to solve contemporary environmental problems

SOCIAL PROFESSIONAL SKILLS

- make meaningful contributions to team work

Course Content

A sound economic analysis of environmental policies is imperative for achieving effective and efficient solutions to solve the important sustainability problems of our planet. Many of these problems arise because the consequences of everyday behaviour on the environment or natural resources are not borne by the individuals who choose how to behave. Market-based policy interventions harness the power of financial incentives to encourage firms and individuals to take into account the full societal impacts of their decisions.

This course begins with foundational concepts in environmental economics, including concepts such as externalities, public goods, and cost-benefit analysis. It introduces key market-based instruments but places stronger emphasis on the practical and political constraints that shape environmental policy. Students will explore how factors such as institutional capacity, political feasibility, lobbying pressure, and information limitations often prevent the adoption of textbook-optimal policies. The course critically examines how these constraints influence the real-world effectiveness, fairness, and efficiency of various policy options.

While covering instruments such as taxes, subsidies, emission trading systems, and individual transferable quotas, the course focuses on how and why governments choose among them in specific contexts. Criteria for policy selection—like administrative simplicity, equity, and resistance to manipulation—are analyzed alongside empirical

case studies. Doing so, the course covers pressing environmental problems such as climate change, the depletion of fish stocks, water management, and air pollution.

Additional Information Teaching Methods

Each week of the course typically consists of two on campus lectures and one on campus interactive. In the lectures we assess the design and use of policy instruments both using traditional criteria (e.g. efficiency, effectiveness) as well as through context-specific constraints (e.g. institutions, transaction costs, public acceptance, distributional concerns etc.). In the interactives we practice with applying our knowledge on policy instruments to specific environmental problems in which these constraints take a central role. In week 6 of the course this application of knowledge is extended to an extensive case study which uses a similar approach. A weekly quiz is used to test understanding of the lecture materials.

Method of Assessment

- Quizzes (10%) (ACADEMIC AND RESEARCH SKILLS , BRIDGING THEORY AND PRACTICE)
- Interactives (active participation) (BRIDGING THEORY AND PRACTICE, SOCIAL PROFESSIONAL SKILLS)
- Group assignment (30%) (ACADEMIC AND RESEARCH SKILLS , BRIDGING THEORY AND PRACTICE, SOCIAL PROFESSIONAL SKILLS)
- Exam (60%) (ACADEMIC AND RESEARCH SKILLS , BRIDGING THEORY AND PRACTICE)

Entry Requirements

Bachelor’s degree in Economics;
Bachelor’s degree in Earth, Economics and Sustainability;

or

Bachelor’s degree in Econometrics, Environmental Sciences, Transport Science, Engineering, or Geography with sufficient background in economics and statistics.

Literature

Title: Policy Instruments for Environmental and Natural Resource Management
Author: Sterner, T. and J. Coria
Edition: 2nd Edition
Publisher: New York: Routledge and RFF Press (2012)
ISBN: 978-1-61726-097-1

Link: https://www.routledge.com/Policy-Instruments-for-Environmental-and-Natural-Resource-Management/Sterner-Coria/p/book/9781617260988?srsltid=AfmBOopcYTq6avkj7_Ac2KP-qJRhk6Vyhlvrfz6YjLV_UJJ2jzxPTwPn

The textbook will be complemented with papers from academic journals.

Energy and Climate Governance

Course Code	AM_1155
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. P.H. Pattberg
Examiner	prof. dr. P.H. Pattberg
Teaching Staff	prof. dr. P.H. Pattberg
Teaching method(s)	Written partial exam, Lecture

Course Objective

After concluding this course, students should be able to:

1. define and explain key concepts of relevance to the climate change governance and energy issue;
2. understand the causes, impacts and effects of climate change and the key scientific controversies in the regime;

3. identify, explain and analyze the various policy options for mitigation and adaptation at different levels of governance with a specific focus on energy related options;
4. understand the key political challenges in the climate change regime, the common problems facing all countries, and the various and changing coalitions in the regime;
5. explain the long-term objective, the principles, the commitments of countries and other key elements of the Climate Change Convention, including the Paris Agreement;
6. to explain the role of energy in the climate change regime, and the various aspects of policy with respect to renewable energy transition;
7. to analyze global energy policy taking into account questions of energy security, energy access and affordability as well as sustainability;
8. to make a judgment about which principles, policy instruments and approaches are likely to be most efficient, equitable and/or effective in addressing the climate change problem, including energy policy.

Course Content

Global governance of human-induced climate change, including both mitigation and adaptation, is a hotly debated subject. Current (international and transnational) climate policy is the result of a complex and long-lasting negotiation process at multiple levels of governance. In this process, the science of the complex earth and climate system is closely linked to questions on the socio-economic effects of climate change, the options for global environmental governance as determined by the structure of international organizations, international economic and political relations and environmental law. Reaching the 1.5 degree temperature target of the 2015 Paris Agreement will require a fundamental transition of global energy systems. Therefore the course will investigate the relation between climate change and global energy policy. The relation between climate change and energy will also take the complexities of the energy trilemma (security, sustainability, equity) into account.

The course includes:

- an overview of the science of climate change, its impacts , uncertainties, mitigation, adaptation;
- discussion of climate change policy options at multiple levels of governance, including the international climate change regime, national and European policies, and transnational approaches;
- analysis of the political challenges in climate change and the positions of different countries and actors;
- assessment of the economics of climate change including analyzing the flexible mechanisms (Emission trading, Clean Development Mechanisms) and options for Post Kyoto measures;
- analysis of interrelated energy challenges (energy trilemma);
- analysis of the challenges for a transition to renewable energy at various scales;
- the relation between global energy policy (including its geo-political aspects) and the climate change regime.

Additional Information Teaching Methods

Seminar/‘Werkcollege’ (s)

Method of Assessment

Written exam (E);
Summary of COP (report)
Simulation (Sim; pass/fail);

Weight of each graded component: 100

Compensation: is it possible to compensate one component with another? NO

Mode of re-examination for the different components: re-sit for final paper

Literature

See Course manual on Canvas

Additional Information Target Audience

master students

Recommended background knowledge

basic knowledge of environmental policy and governance

Environmental Economics for ERM

Course Code	AM_468020

Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. P.J. Robinson
Examiner	dr. P.J. Robinson
Teaching Staff	dr. P.J. Robinson, dr. ir. M.A. Poblete Cazenave
Teaching method(s)	Seminar, Study Group, Lecture

Course Objective

This module gives an introduction to the economic analysis of environmental problems, and aims to give an overview of economic environmental policy instruments in different policy contexts from the local through to the global level. A critical cause of environmental problems is that not all of the costs (including environmental) caused by economic agents are borne by those responsible for generating them.

This problem will be conceptualized in this course through the notion of externalities. There are various instruments and institutional arrangements for addressing such externalities. Criteria for their selection and evaluation will be studied. Applications of environmental policies at various administrative levels (i.e. local, national, international), different economic sectors and different country contexts will be discussed. This course tackles some of the most pressing environmental problems that our planet faces today, such as the depletion of common pool resources, climate change, and environmental effects of trade and globalization.

The overarching objective of this course is to familiarize students with the economic analysis of environmental problems. After following this course, students should be able to judge how well certain policy instruments and institutional arrangements perform in terms of effectiveness, efficiency and the distribution of welfare in society.

Course Content

After having participated in this module, students should have the knowledge to do the following:

1. Describe the fundamental nature of environmental problems from an economic perspective, in relation to notions like externalities, public goods and free riding.
2. Derive optimal levels of pollution and resource use from a societal perspective.
3. Describe what these economic fundamentals and optima imply for the feasibility of various solutions to existing environmental problems.
4. List economic policy instruments that are available, and discuss their (dis)advantages with respect to predefined policy aims.
5. Argue how international trade, poverty and the environment are interlinked, and discuss the role of trade in stimulating or hampering sustainable development.
6. Describe what are critical and debatable assumptions of core policy insights within environmental economics (e.g. those related to economic costs-benefit analysis of climate policy).

In this course "Environmental Economics", one distinct subject will be addressed per week. This implies that various activities (e.g. (guest) lectures, interactive events) will take place on a weekly basis addressing a distinct central topic.

Additional Information Teaching Methods

- Lectures
- Reading
- Quizzes
- Q&A sessions and debates
- Serious gaming
- Tutorials/consulting hours

Method of Assessment

Last year's grade was determined as follows:

- Individual closed-book exam;
- 3 group assignments.

Students must pass both elements (5.5 or higher).

Entry Requirements

This course is suited for students with a broad range of disciplinary backgrounds. A background in economics is no prerequisite.

Literature

Harris, J.M. & Roach B. (2018). Environmental and Natural Resource Economics: A Contemporary Approach – Fourth Edition. Routledge: Taylor & Francis Group, London.

Additional Information Target Audience

This course is suited for students with a broad range of disciplinary backgrounds. This course provides some of the fundamental building blocks for the MSc Environmental and Resource Management (ERM). It provides the basic framework for economic research methods, such as cost-benefit analysis and environmental valuation. We encourage students who are more interested in thorough theoretical/ mathematical analysis to follow the environmental economics course provided by the School of Business and Economics.

Additional Information

Main lecturers/co-coordinators for this course are Dr. Miguel Poblete and Dr. Peter Robinson, the teaching assistant is MSc Zsóka Halászová. This course provides a sound balance between theoretical lectures and guest lectures by academic and policy experts in the field of environmental economics. The interactive sessions, serious gaming events, quizzes and active reading aim to stimulate active learning of students. To stress the societal importance of environmental economics, guest lecturers in this course are also from outside academia.

Environmental Policy

Course Code	AM_468021
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. Blondeel
Examiner	dr. M.C. Blondeel
Teaching Staff	dr. O.E. Widerberg, C.E. Rappel
Teaching method(s)	Seminar, Lecture

Course Objective

The Environmental Policy course introduces students to the actors, institutions and instruments relevant for designing and implementing environmental policies at the local, national and supranational levels. It also introduces various evaluation techniques for analyzing the environmental policy in an increasingly complex world.

After this course, students should be able to:

1. Describe the public and private actors and institutions involved in environmental governance at local, national and supranational levels.
2. Evaluate the effectiveness and legitimacy of environmental governance at local, national and supranational levels.
3. Explain, analyze and propose solutions regarding the complexity of decision-making structures, procedures and negotiations of environmental issues.
4. Communicate the acquired knowledge from the course on the basis of a video assignment and essays.

Course Content

This course introduces environmental policy from a political science, public policy, and international relations perspective. Environmental policy is understood in terms of governance, going beyond rules and regulations by national governments. Students will learn the basic theories and concepts in environmental policy at the local, national and international level. It also discusses interaction between the various levels of governance. Besides theory and concepts, the course provides students with practical analytical tools for analyzing the effectiveness and legitimacy of environmental policies that can be applied in as well as outside academia.

The course is structured around three modes of governance: Hierarchy (command-and-control instruments, primarily carried out by public or governmental actors); Markets (market-based governance, primarily market-based instruments); and Networks (the functioning and effects of partnerships, labeling initiatives, and actor networks

(with a focus on non-state actors)).

The final week is dedicated to an emerging topic in environmental policy that changes every year, for instance, environmental activism, animal rights, transformations, AI, or degrowth.

Additional Information Teaching Methods

The course is worth 6 ECTS points which corresponds to 150 hours of work per student. The course is based on a series of lectures, exercises and debates.

Method of Assessment

1. Group video assignment (40%)
2. Individual papers (60 %)

Literature

Please see syllabus on Canvas.

Additional Information Target Audience

Master students

Food Governance & Sustainability

Course Code	AM_1263
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K.E. Negacz
Examiner	dr. K.E. Negacz
Teaching Staff	dr. M.J. Koetse, dr. K.E. Negacz
Teaching method(s)	Seminar, Lecture

Course Objective

To provide students with a profound understanding of how the public and private sector, as well as consumers can contribute to the transformation towards a more sustainability food system, and to challenge students to design their own solutions contributing to this transformation.

By the end of this course, students should:

1. Understand how barriers related to policy, regulation, finance, innovation, or behaviour lead to unsustainable food systems;
2. Learn about how nature-based solutions can be designed to support sustainable food producers;
3. Understand the challenges of food justice and food sovereignty;

Students should be able to:

1. Examine how policies and interventions are implemented to diversify food supply and reduce food waste;
2. Analyse opportunities in the food supply chain are created through digitalization, investments & gender-specific measures;
3. Critically examine how transformation will look different from one country to the next, and from one food system to the next;
4. Synthesize and evaluate the academic literature and lessons learned during the course for a specific food transition process at a certain governance level (international, national, regional, local) into a pitch for experts.
5. Feel confident to work efficiently in groups and take feedback on board for individual assignments.

Course Content

The world faces an enormous challenge to increase the sustainability of the current food system thereby helping to ensure healthier diets, increase food security, and support rural communities while addressing climate change and safeguarding biological diversity. Alternatively, leaving food systems to follow current trends will further jeopardize human health, biodiversity, climate, natural resources, and rural economies. Therefore, transformation of food

systems is an urgent priority for the public and private sector, civil society, international organizations, research communities, and consumers.

Can a sugar or fat tax drastically influence consumers food choices? How can we encourage farmers to grow crops and cattle more sustainably? And are there ways to allocate profits more fairly across the food supply chain? To be able to answer these questions a thorough understanding of the governance of sustainable food systems is required, ranging from local to global interventions.

The master course “Sustainable Food Governance” aims to provide students with a profound understanding of how the above-mentioned stakeholders can contribute in their own way to the transformation towards a more sustainability food system. In doing so, we adopt a holistic agenda developed by leading international experts which is centered around ten critical transitions that would enable food systems “to provide food security and healthy diets for a global population of over nine billion by 2050, while also tackling our core climate, biodiversity, health and poverty challenges” (Food and Land Use Coalition – FOLU 2019).

Additional Information Teaching Methods

This course will follow the principle of blended learning, which is an educational approach that combines online educational materials and opportunities for interaction online with traditional place-based classroom methods, offering flexibility for both teachers and students. In total the course will consist of main live and online lectures, working group sessions, and online and live guest lectures. Students will develop their own proposition on how to contribute to the transformation towards a more sustainable food system.

Method of Assessment

The course is assessed through:

- Written exam (50%)
- Pitch presentation (40%)
- Paper club presentation (10%)

Students must pass all elements (5.5 or higher).

Entry Requirements

Students do not require a specific disciplinary background, although affinity with food sustainability and food security is useful. An emphasis within this course is the use of ‘systems thinking’ which is not a discipline, but a way of thinking about problems that tries to be holistic, focus on interdependencies and dynamics, and the different people involved.

Literature

- Jennifer Clapp (2020) Food - third edition. Polity Press. Cambridge. ISBN-13: 978-1-5095-4177-5
- Further readings will be available on Canvas.

Food Systems & Sustainability

Course Code	AM_1262
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Torralba Viorreta
Examiner	dr. M. Torralba Viorreta
Teaching Staff	C.E. Rappel, dr. ir. C.J.E. Schulp, dr. M. Torralba Viorreta, prof. dr. ir. P.H. Verburg
Teaching method(s)	Seminar, Lecture

Course Objective

To provide students with a profound understanding of food systems and their dynamics, how different elements of the food system are connected, and opportunities for the transformation towards a more sustainable food system. By the end of this course, students are able to understand:

1. existing concepts available to represent food systems;

2. the components of food systems and their interactions;
3. the role of food production through land use;
4. the role of diets and consumption in food systems;
5. the role of value-chains in the food system;
6. how food systems connect producing and consuming regions, often over large distances;
7. different methods and approaches for assessing and evaluating food system sustainability.

and have gained the skills to

1. Apply theoretical food system knowledge gained throughout the course.
2. Analyze food system-related data to assess sustainability outcomes.
3. Evaluate solutions for improving sustainability in food systems using real-world examples.

Course Content

Food production leads to major global environmental change impacts and with changing consumption patterns these impacts are expected to further increase in the near future. The world faces a massive challenge to increase the sustainability of the current food system, thereby helping to ensure healthier diets, increase food security, and support rural communities while addressing climate change and safeguarding biological diversity. To be able to address this enormous challenge, a thorough understanding of the functioning of food systems is required, ranging from the production of food to the choices underlying consumption patterns and the value chains that connect these.

The master course “Food Systems & Sustainability” aims to provide students with a profound understanding of the functioning of food systems and the ways these can be studied and evaluated in terms of sustainability. We will discuss the different components of a food system and focus on the interactions and feedbacks in the food system.

Special attention will be paid to the ways in which the food system affects multiple sustainability dimensions.

Besides aiming at an understanding of the food system as a whole, the course will address the ways in which food systems vary across the world and how they change with time, ranging from local subsistence systems to globally connected value-chains governed by mega-companies. Students will be introduced to different scientific methods to analyze food systems and evaluate their sustainability implications.

Additional Information Teaching Methods

- Lectures (HC); 10 sessions of 90 minutes - Working lectures / seminars (WC); 2 sessions of 90 minutes

Method of Assessment

- Closed book exam: 60%.
- 3 assignments (individual or small group; exercises + short reports): 40%

Students must pass all elements (5.5 or higher) in order to pass the course.

Entry Requirements

Students do not require a specific disciplinary background, although affinity with food sustainability and food security is useful. An emphasis within this course is the use of 'systems thinking' which is not a discipline, but a way of thinking about problems that tries to be holistic, focus on interdependencies and dynamics, and the different people involved.

Literature

The course will use a syllabus that will be made available to the students. In addition, several journal articles and chapters from scientific reports are part of the reading material and will be made available via Canvas.

Additional Information Target Audience

Students with an interest in the link between the natural environment and society and in sustainability problems of global food systems.

Explanation Canvas

All course materials, assignments, and quizzes will be made available via Canvas only.

Recommended background knowledge

Students will benefit from some experience with quantitative data and should be able to conduct data analysis in

Excel.

Governance of Ecosystem Services

Course Code	AM_468025
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Schaafsma
Examiner	dr. M. Schaafsma
Teaching Staff	dr. M. Schaafsma
Teaching method(s)	Seminar, Lecture

Course Objective

The aim of this course is for students to acquire profound knowledge and understanding of the governance of ecosystems and biodiversity, specifically to:

1. understand the concepts and key elements of ecosystems and biodiversity governance
2. analyse governance issues and evaluate the strengths and weaknesses of various policy instruments and organizational frameworks related to sustainable ecosystem services and biodiversity management in different contexts.

After following this course, students should be able to:

- discuss the characteristics of ecosystems and biodiversity and how these relate to governance mechanisms worldwide (LO1);
- discuss available policy instruments at different governance levels (LO2);
- analyse the legitimacy, transparency, accountability, inclusiveness, fairness, integration, capability, adaptability, and effectiveness of these governance options in providing ecosystems and biodiversity protection (LO3);
- discuss potential trade-offs between the protection of ecosystem services and biodiversity and other socio-economic interests at the local community, regional or national sector level (LO4).

Course Content

Current governance systems of ecosystem services and biodiversity have evolved through policies developed at local (e.g. community based forest management), national (e.g. protection of wetlands), supranational (e.g. EU legislation such as Natura2000) and global levels (e.g. Convention on Biological Diversity). Governing biodiversity and ecosystem services is complicated because of a number of factors, such as their public good characteristics, different value systems, the often unclear distribution of user and property rights, the trade-offs between resource use and other societal goals, distributional outcomes, lack of coordination between various actors, and high transaction costs of effective enforcement and control. Also, scientific uncertainty surrounding the relationship between biodiversity protection, ecosystems and ecosystem services as well as the complexity of how socio-ecological systems function and change, including the presence of thresholds, play a role. Yet, because of the crucial nature of ecosystem services and biodiversity in human survival, ‘good’ governance of ecosystems and biodiversity is highly important.

The course will focus on how ecosystems are currently managed, what policies and policy measures are in place, and what lessons have been learned so far in developing and implementing different governance structures and policy instruments at the local, national and international level to safeguard ecosystems and biodiversity in the long run.

The course will elaborate on the multi-dimensional aspects of the governance of ecosystems and biodiversity, illustrated for different types of ecosystems and governance structures, varying from local community-based approaches to Payments for Ecosystem Services (PES) and global agreements and coordination such as the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES).

Important governance questions that hence arise in practical ecosystem and biodiversity governance and management, and that will be discussed during the lectures, are:

- What are the key institutions and who are the actors? Who is included in governance processes? Who holds power?
- How do the actors interact? What are the processes? How are decisions made?

- What are the links between the different levels of decision-making (local, national, global)?
- What policy instruments are used? What type of interventions are discussed?
- How do the policies, instruments and/or interventions score on different governance evaluation criteria?

These questions also form the type of questions to consider for the paper.

Additional Information Teaching Methods

The course corresponds to 6 ECTS and consists of a combination of (guest) lectures and interactive sessions over a period of 4 weeks, required reading, a paper and presentation, and an exam. Various topics concerning the governance of ecosystem services will be discussed during the lectures, guest lectures and interactive sessions. Literature related to the (guest) lecturers' presentations is provided on Canvas. The course involves multiple guest lecturers, who translate theories and concepts of ecosystem services management and governance into policies and actions. The guest lectures will be supported by literature on governance of ecosystems and theory lectures.

In addition to the (guest) lectures, students will work on a short scientific paper about a topic of their own choice. A list of suggested topics to choose from will be provided at the beginning of the course. To guide students in their writing, an interactive session is organized once per week (also on the first day of the course). The paper is developed and written during the first 3 weeks of the course and presented in the fourth week. The paper and the presentations are compulsory and will be graded.

Classes are scheduled in the first three weeks of the course and, together with the required reading, paper writing, presentation and exam preparation, are estimated to account for about 168 hours of students' time. Attendance in all classes is mandatory, but will be checked irregularly. We assume students are sufficiently serious and will show up at every class, barring unforeseen circumstances. Attending class is likely to increase your grades for the exam and the paper, so attending classes is in your own interest. Importantly, we all bring different expertise, experiences, questions and perspectives to the online classroom, so please share and learn – this will make the course more interesting and fun.

Indicative division of course hours:

- Class attendance: 33 hours
- Mandatory readings: 35 hours
- Writing paper and preparing presentation: 68 hours
- Preparing and taking the exam: 32 hours
- TOTAL: 168 hours (Standard total time allocated for 6 ECTS).

Method of Assessment

The overall mark for this course is based on three separate grades, i.e.:

- Closed-book written exam (Individual; 40%);
- Short Scientific Paper (50%);
- Presentation (10%).

A necessary condition to pass this course is that all three grades must be 5.5 or higher. The exam has a resit. If a student obtains a grade lower than 5.5 for the paper or the presentation, an assessment will be made to determine how the student can improve their grade.

Further details about the assignments and examination will be provided in the course manual.

Literature

See the course manual on Canvas. The compulsory reading list will contain about 15 academic papers which will be closely related to the topics of the guest lecturers (to be read in advance).

Additional Information Target Audience

MSc students Environmental Resource Management, MSc students Earth-Sciences/Global Environmental Change and Policy track (doctoral students or other MSc students are also welcome: please contact the course coordinator).

Recommended background knowledge

Students are encouraged to take the courses 'Value of Ecosystem Services and biodiversity' (AM_468024) and 'Environmental Policy' (AM_468021).

Methods of Environment and Resource Management

Course Code	AM_1135
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	A.H. Levering
Examiner	A.H. Levering
Teaching Staff	dr. M. Tesselaar, A.H. Levering, dr. ir. C.J.E. Schulp, dr. M.C. Blondeel, dr. K.E. Negacz
Teaching method(s)	Excursion, Study Group, Seminar, Computer lab, Written partial exam, Lecture

Course Objective

The objective of the course is to provide students with an overview of relevant scientific methods for studying sustainability problems. After the course, students should be able to:

1. Understand and explain the main methods for studying sustainability problems;
2. Apply methods for studying sustainability problems;
3. Interpret, analyze, and critically reflect on the results gained from applying the methods in steps 1 and 2.

Course Content

The course provides students with a hands-on introduction to four analytical methods used to study sustainability problems: economic analysis, scenario analysis, spatial analysis, and stakeholder analysis. All methods are offered through a dedicated module, and students need to choose two out of the four modules. The methods will be taught through a combination of lectures, practical classes, and (computer) exercises. The methods will be applied on (simplified) sustainability problems, and students will reflect on the results.

Additional Information Teaching Methods

The methods will be taught through a combination of lectures, practical classes, and (computer) exercises. The methods will be applied to current sustainability problems, and students will reflect on the results.

Method of Assessment

The assessment and resit procedure for this course are as follows:

1. Each module will be evaluated separately, and the score for this course will be calculated as the average score of the two selected modules.
2. Each module will be evaluated based on either individual or group assignments, depending on the module. Details on these assignments will be provided on Canvas.
3. Students must receive at least a 5.5 for each module separately. This means that compensation between modules to receive an overall passing grade is not possible.
4. If a resit is taken for a module, then the maximum grade for that module that can be achieved is capped at 6.5. The overall grade for the course can still exceed 6.5 when only one of the two modules requires a resit.

Literature

See the study manual on Canvas

Custom Course Registration

Students will register for this course and choose two out of four modules in the weeks leading up to the start of the course, in consultation with their thesis supervisor. The modules are chosen through Canvas, after signing up for the course.

Research Design

Course Code	AM_1136
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	dr. I.M. Lehmann
Examiner	dr. I.M. Lehmann
Teaching Staff	dr. J.F. Brusselaers MA, P. Robertson MSc BA, dr. I.M. Lehmann, dr. G.A. Dreschler, dr. M. Mazzoleni
Teaching method(s)	Seminar, Study Group, Lecture

Course Objective

The goal of this course is to prepare students for their Research Project (i.e., their MSc thesis), which starts immediately after completion of the Research Design course. At the beginning of the course, students will already know the topic of their MSc thesis, and the course will help them to write a research plan for implementing this topic. This means, the course will guide them through the different stages of developing a research plan from the literature review and specification of the research question over identifying appropriate theories and methods to setting up a time plan for the thesis. At the end of the course, students will have written their own research plan and prepared a short presentation to pitch it. The course is accompanied by an extra independent module on academic English writing.

At the end of the course,

- Students have gained in-depth understanding of a societal problem and the related state-of-the-art of research in a selected field of environment and resource management
- Students are able to formulate a concise and novel research question so that answering it will identify solutions to the societal problems and contribute to academic knowledge
- Students are able to develop a research design to answer their research question, which includes being able to choose appropriate concepts or theories, methods and material
- Students can effectively plan a long-term research project.
- Students can synthesize and communicate their research plan in written and oral forms at academic level.

Course Content

The course begins with an introductory lecture for all students, which explains the purpose of a research proposal and its key components and explains the logic and requirements of the course. This is followed by weekly parallel workshop-style tutorials. Each tutorial will cover the same content and will be organized in the same way: Tutors will provide additional input in the form of lectures and informal discussions on how to produce a research design for an MSc thesis and how to improve important soft skills. Students will moreover have room to work on their proposals during the tutorials and give each other peer feedback. Weekly homework related to the different elements of a research proposal will help students to stay on track and develop their proposals in a step-by-step approach. The complementary academic writing module will give students the opportunity to work on the structure and style of their text.

Additional Information Teaching Methods

Seminar/‘Werkcollege’ (s)

Method of Assessment

The grades for the Research Design course consist of three elements:

1. A written Research Plan for the Research Project (70%, graded by thesis supervisor)
2. (Recorded) presentation of the Research Plan (30%, graded by the course tutor)
3. Students need to receive a "pass" grade from the Academic Writing course (graded by Academic Writing teachers).

Should a student fail the course, we offer a resit for the presentation and a revised version of the research proposal as well as the Academic Writing component. We advise students that expectations of tutors and supervisors may increase given the extra time available in the course of the resit period.

Literature

See Course manual on Canvas

Research Project ERM

Course Code	AM_468017

Credits	18

Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.F. Brusselaers MA
Examiner	dr. J.F. Brusselaers MA
Teaching Staff	dr. J.F. Brusselaers MA
Teaching method(s)	Seminar

Course Objective

After completing the research project, the student should be able to:

1. further develop and apply the key concepts, theories and methodologies taught in previous ERM courses in a specific environmental or resource context;
2. independently develop and execute scientific research of societal relevance;
3. reflect on matters and issues in the chosen domain of environmental and resource management, and is able to form an opinion and contribute to both scientific and practitioner's discussions;.
4. present and discuss research findings in a meeting with peers;
5. independently write a report on the research and its findings in an academic style.

A research project is carried out individually and independently under supervision of an appointed supervisor and after approval of the research project Work Plan by the supervisor.

Course Content

In the Research Project (RP) course of the ERM programme, the student will individually carry out a research project at one of the research groups at the VU University Amsterdam or externally at another university, (inter-)governmental organisation, a non-governmental organisation, or company. All non-VU organisations are hereafter referred to as external organisation. The study credits for the course are 18 ECTS, which equals 63 days of work full-time (i.e. 8 hours a day).

The student identifies the subject of his or her RP before the start of the Research Design course preceding the RP course, including a suitable supervisor and (optional) placement, which fits the interests of the student and, in case of an external placement, those of the external organization.

The collaboration between student and supervisor is formalized in an Agreement Form. During the Research Design course the Work Plan of the RP is written. Like the Agreement Form, this Work Plan is submitted online via Onstage to the RP supervisor for approval before the start of the RP.

Additional Information Teaching Methods

During the research project, communication and information exchange between the student and supervisor is paramount to ensure the RP develops according to plan. This may happen via in person consultations, e-mail, or online meetings. When you prepare your Work Plan you should agree with the supervisor and make explicit how and how often you will communicate and exchange information about your progress with your supervisor. Make sure that both you and your supervisor are satisfied with the arrangement and that you, as a student, are confident that you will get enough feedback to do your research properly and in time. To ensure that meetings with your supervisor is efficient , it may help to have a clear agenda, prepare the subject(s) you wish to discuss in advance and make a short report of the meeting (definition of actions and decisions taken, appointment for next meeting) and send this to your supervisor after the meeting. This is especially important when you are going abroad for a certain part of the RP.

Method of Assessment

The ERM RP and the final report will be assessed by both the supervisor and an independent second assessor via an online Assessment Form. The second assessor is appointed by the RP coordinator. The assessment follows a double blinded peer review procedure. This means that neither student nor supervisor will know beforehand the identity of the second assessor. The assessment of the RP will be based on the following three components:

Final Report (80% of the total grade based upon five categories of criteria):

a) Introduction and framing of the study (15%)
The extent to which the problem that the study tries to address is clearly described, the research is interdisciplinary and clearly positioned within the existing research on the topic, the research question is clear, relevant and

appropriate given the problem statement, and the novelty and originality of the research question and/or research methodology, is clearly defined and justified.

b) Methodology of the study (20%)
The extent to which the chosen research methodology is appropriate to the research problem at hand, the research method is well described, and correctly applied.

c) Presentation of results and discussion (35%)
The extent to which the report clearly and correctly conveys the research results to the reader, the tables and figures are of sufficient quality, the information was gathered from a variety of sources and was critically scrutinized, shows sensitivity to the complexity of the research, and appropriately deals with assumptions and uncertainties.

d) Conclusions and abstract (15%)
The extent to which the conclusions answer the research question (and sub- questions), are supported by the research, grasp the main points and provide a clear take-home message, and give useful recommendations (scientific and/or societal). The abstract is informative and conforms to the rules.

e) Coherence and formatting (15%)
The extent to which the report is logically structured, in-text citations and the bibliography are consistent, complete and in proper format, the language is appropriate for a scientific report, and grammar, spelling and punctuation are correct.

The final report is evaluated by the student's supervisor and an independent second assessor based on the criteria above.

2.Execution (10% of total grade)

- Degree of independence with which the student carried out the RP
- Communication with supervisor: keeping appointments and handling of feedback
- Time management

The way the student executed the RP is evaluated by the student's supervisor based on the criteria above.

3.Oral presentation and defence (10% of total grade)

- Content: clear, understandable story line
- Narrative techniques: captivating way of presenting (voice, gestures, involving audience)
- Use of audio-visual devices (Power Point, Film)
- Debating capacities (handling of questions)
- Time management (within the allocated 15 minutes).
- The presentation is evaluated by the student's supervisor based on the criteria above.

Note: A no-show of the student for the scheduled presentation, without prior notice and/or valid reason, means a fail (grade 0). An attendance list will be used to check if all students attend the presentations of their fellow students. Attendance is mandatory.

4.Attitude (insufficient/sufficient/good)

This criterion has to be sufficient in order to get a grade and refers to a student's general work attitude in the RP, motivation and commitment. The student's attitude is evaluated by the supervisor based on the student's performance during the entire RP period.

The assessment of the RP is partly based on the final report (80%), execution (10%) and presentation (10%). For all three components the student has to have a score of at least 6.0. Also the attitude has to be sufficient in order to be able to pass the course. Numerical grades for the different components are given using one decimal point. The final grade is rounded to a half or whole decimal point.

The grading of the final report is based on the five criteria mentioned above. Here no minimum limits to the grading apply as long as the overall grade is 6.0 or higher. Presentation of the results and discussion is considered most important and is therefore given the highest weight (35%), followed by methodology (20%), and introduction and framing of the study (15%), conclusions and abstract (15%), and coherence and formatting (15%).

The final grade consists of a weighted average of the grades given by the supervisor and the independent second assessor. The second assessor only evaluates the final report based on the same criteria as the supervisor. Both have to grade the final report as 6.0 or higher. The assessment of the final report by the supervisor is given more weight (70%) than the assessment of the second assessor (30%) in view of the fact that the supervisor has more in-depth knowledge and information about the student's abilities and the writing of the final report.

If one of the two grades the final report lower than 6.0 or the difference in grading is more than 2 points, a third assessor is appointed by the RP coordinator following the same double blinded review procedure and asked to

evaluate the final report. The grade of the third assessor is in that case decisive in the final outcome of the assessment. Hence, if also the third assessor grades the final report lower than 6.0, this results in an overall fail for the RP. If the grading by supervisor and second assessor was higher than 6.0 but resulted in a difference of more than 2 points, the final grade is established by taking the mean of the grades given by the second and third assessor and giving this a weight of 30% in the final grade.

Entry Requirements

To be allowed to start the ERM Research Projects course, students must have participated in the courses Research Methods (AM_1135) and Research Design (AM_1136) and obtained a minimum of 18 ECTS in the ERM.

Literature

The literature review is depending on the subject and the option chosen.

Additional Information

The Research Placement is subject to the FALW Work Placement and Thesis Regulations. These regulations require detailed written agreements between supervisor and student that specify the conditions for the Research Project. Please consult the Faculty's website for more information.

Recommended background knowledge

Students are strongly recommended to have followed: Environmental Economics (AM_468020) and Environmental Policy (AM_468021)

Sustainable Energy Systems

Course Code	AM_468018
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. M.A. Poblete Cazenave
Examiner	
Teaching Staff	dr. M. Werthschulte, dr. ir. M.A. Poblete Cazenave
Teaching method(s)	Study Group, Lecture

Course Objective

The goal of the course “Sustainable Energy Systems” is to introduce students to key aspects of the energy transition from a systemic socio-economic perspective. Besides a brief introduction to the technological aspects of the energy transition, the lectures will focus on the economic, societal, and political conditions for sustainable energy generation, distribution, and use. The applied component of the course will stimulate the students to critically reflect on the positions of various stakeholders in the energy transition and to develop their own ideas for business, community-oriented projects or public policies that could accelerate the energy transition.

After following this course, students should be able to:

1. Understand the basic technical and economic principles underlying today’s energy generation, distribution, and consumption around the globe and have an idea of the possible pathways into a sustainable energy future.
2. Understand the economic, social, and environmental challenges related to the energy transition and critically discuss possible solutions and strategies for the decarbonization challenge.
3. Know about a diverse set of policy instruments and strategies that can be implemented to support the energy transition.
4. Think critically about efficiency, effectiveness, fairness, and distributional consequences of different principles, policy instruments, and approaches to the energy transition. Learn to recognise trade-offs and potentially competing objectives.
5. Identify niches for sustainable energy innovations, understand the main components of a successful business model or community project and apply this in developing a business plan or community project for a clean energy innovation.
6. Understand stakeholders’ positions and how to constructively debate to find synergistic solutions.
7. Demonstrate the capacity to operate in a multidisciplinary team and constructively contribute to the

development of a shared project.
8. Synthesize information into a pitch for a non-academic audience.

Course Content

Environmental externalities like CO2 emissions leading to global climate change and air pollution urge the world to transition towards a more sustainable and less-carbon intensive renewables-based energy system. Energy is central to nearly every major challenge and opportunity the world faces today, such as health, employment, education, or food production. Furthermore, access to clean and sustainable energy for all is essential and part of the UN Sustainable Development Goals (UN, 2015). The transition towards an affordable, reliable, and sustainable energy system can be accelerated by investing in renewable energy sources, prioritizing energy efficient technologies and practices, and by building a clean energy infrastructure, but changes to the physical infrastructure need to go hand in hand with economic, societal, and behavioural changes.

The course "Sustainable Energy Systems" is built on a series of lectures and seminars by VU lecturers and selected guest speakers to introduce students to key aspects of the energy transition from a socio-economic and socio-technical perspective. Every lecture will be a combination of theory and evidence-based discussions, relating theoretical arguments with recent experiences and examples in the domain of the transition to sustainable energy.

The course is built on 4 main blocks:

1. The socio-technical systems perspective of the energy transition (socio-technical pathways for greening energy production, distribution and use)
2. Sustainable energy demand innovations and deployment: from available energy efficient technologies to public programmes and policies to international agreements
3. Energy markets and policies (economics and policy instruments to support the energy transition, future electricity markets)
4. Consumer behaviour and social aspects (sustainable energy consumer behaviour, energy poverty, energy access)

Additional Information Teaching Methods

The course is worth 6 ECTS credits which corresponds to 168 hours of work per student. The course comprises two sets of activities. The first takes place in class, where information is presented through lectures, presentations, debates, seminars, etc. The second includes assessments where the student’s ability to achieve the course’s objectives is tested. Assessments involve group assignments (presentation and written assignments), and the exam. Feedback opportunities are included through meetings as well as assessments. Approximate time allocation: • Class: 28 hours • Reading and exam preparation: 70 hours • Assignment: 60 hours • Presentations: 10 hours.

Method of Assessment

Type of assessment:

- Two group written assignments (A) worth 30% of the final grade.
- One group project involving a written assignment (A) and presentation (Pres) worth 40% of the final grade.
- Written exam (E) worth 30% of the final grade.

Minimum grade to pass the course: 5.5 in each assessment component. It is not possible to compensate one component with another. It is compulsory to attend the group presentation session.

Value of Ecosystem Services

Course Code	AM_468024
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.J. Koetse
Examiner	dr. M.J. Koetse
Teaching Staff	dr. F. Riva MSc, dr. G.V. Farina, dr. M. Schaafsma, dr. M.J. Koetse, dr. M. Torralba Viorreta
Teaching method(s)	Excursion, Seminar, Study Group, Lecture

Course Objective

One of the hot topics in environmental management today is the concept of "ecosystem services". This concept can be defined as the benefits that people obtain from natural ecosystems. By putting ecosystem services central in the debate on nature and biodiversity conservation, the societal benefits of conservation and the need to align conservation and development goals are stressed. Quantification of biodiversity and ecosystem services, and identifying the beneficiaries and their associated values, are major challenges in this respect. This course aims to assess the importance of ecosystem services and biodiversity for nature, the economy and people.

Course Content

The course focuses on an understanding of what specific services are provided by ecosystems, and how these are linked to biodiversity. It also extensively addresses the economics of ecosystem services, the use of (economic) valuation methods to measure these values, and the use of decision support tools to structure and inform decision making. Various ecosystems (e.g. forests, coral reefs, wetlands, rivers) and forms of biodiversity are covered (e.g. wildlife, flora). Next to main and guest lectures and debates, the course contains a six-week case study carried out in groups of 3-4 students. Potential subjects addressed in this group assignment will be provided by the main lecturer, and are generally linked to ongoing research projects. Topics may also be developed by the students themselves. The main objective of the assignment is to apply the skills taught in the course in a real-life example, and to develop skills in academic writing, effective communication and working in a diverse group.

After having participated in this module, students should be able to do the following:

1. Understand how ecosystems produce ecosystem services and how ecosystems change over time and space.
2. Describe how biodiversity relates to ecosystem services, what services ecosystems and biodiversity provide, and how these services can be quantified and mapped.
3. Discuss the social and economic importance of ecosystem services and biodiversity and what drives the social and economic values.
4. Understand the main characteristics and elements of different (economic) valuation methods and their applications.
5. Understand and discuss the differences between classic economic monetary valuation, and alternative approaches to valuation and values.
6. Discuss decision making tools and methods for appraisal of ecosystem services, from both a public and a private sector perspective, and conduct an appraisal.
7. Understand and discuss differences and similarities between different decision support tools.

Additional Information Teaching Methods

- General lectures
- Prerecorded clips and quizzes
- Debates
- Group paper and discussions
- Peer review

Method of Assessment

- Closed book exam (T: 60%).
- Group paper and paper presentation (V+Pres: 40%);

Students must pass both elements (5.5 or higher).

Entry Requirements

Students do not require a specific disciplinary background, although affinity with ecology and economics is useful.

Literature

The main written source is a book (i.e. Bouma and Van Beukering 2015) which is written especially for this specialization in Ecosystem Services and Biodiversity.

Bouma, J.A. and P.J.H. van Beukering (2015). Ecosystem services: from concept to practice. Cambridge University Press. Cambridge.

In addition, several freely downloadable journal articles and papers are part of the reading material.

Additional Information Target Audience

Students with an interest in the link between the natural environment and society. This interest may vary widely,

possibly including topics such as nature conservation, tourism, food production, poverty, international trade, education, and landscapes.

Additional Information

The course is coordinated by Dr. Marije Schaafsma and Dr. Mark Koetse. Teachers involved are Dr. Marije Schaafsma, Dr. Mark Koetse, Dr. Georges Farina, Dr. Mario Torralba, and Dr. Federico Riva. This course may also include various guest lectures by experts and practitioners in relevant fields.

Water extremes: Risks and Solutions

Course Code	AM_1305
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Mazzoleni
Examiner	dr. M. Mazzoleni
Teaching Staff	dr. P. Scussolini, dr. M. Mazzoleni, dr. M.L.K. Wens, prof. dr. J.C.J.H. Aerts, dr. F.M. Béen
Teaching method(s)	Computer lab, Lecture

Course Objective

The primary objective of this course is to equip students with a comprehensive understanding of the various types of water extremes, risk assessment methodologies, their causes and impacts, and mitigation and adaptation solutions. The course aims to blend theoretical knowledge with practical applications, emphasizing climate change's role and the need for sustainable, community-based solutions. This course provides students with a multi-disciplinary view of water-related disaster risk reduction, with specific focus on floods and droughts.

After finishing this course, the student should be able to understand, analyze, and evaluate:

- Water Extremes assessment: To provide students with a thorough understanding of the various types of water extremes, focusing on floods and droughts. This includes exploring the relationship between climate change and water extremes, including how long-term trends (e.g., climate and land-use changes) are expected to influence the frequency and intensity of these extremes.
- Risk management: To familiarize students with the tools and methodologies used in risk assessment and management, including hazard mapping, vulnerability assessment, and risk evaluation.
- Impact assessment to understand and discuss the impacts of water extremes on different sectors (e.g. water quality and human health). This includes understanding the physical processes that lead to these disasters and their impacts on society.
- Mitigation and Adaptation Solutions: To discuss various mitigation and adaptation strategies for water extremes, including both structural measures (e.g., dams, levees) and non-structural measures (e.g., land-use planning, early warning systems).
- Human-water dynamics: Understand the complex interactions between various water-related issues (e.g., scarcity, floods, pollution), climate, the environment, and the socio-economy

Course Content

Water extremes, such as floods and droughts, are increasingly affecting millions of people worldwide. This trend is likely to continue in the future due to the negative impacts of climatic and socio-economic changes. Floods and droughts are not merely natural hazards, as humans have significantly altered most river basins around the world thereby leading to alterations of the hydrological regime. For example, hydrological droughts can be substantially enhanced by irrigation and groundwater exploitation, while the construction of levee systems can significantly change the spatio-temporal distribution of floodings. As such, to understand how floods and droughts risk changes over time, we need to explicitly consider the complex dynamics generated by the interplay of humans and (both) physical drivers. Different management strategies have been adopted to reduce the negative impacts of floods and droughts. Structural (e.g. levee system, reservoir constriction) and non-structural measures (e.g. water use reduction, early warning systems), have been extensively used over the last decades. However, structural measures that address the problem in the short term often generate unintended consequences in the long term because of the feedback mechanisms operating in coupled human-water systems. For these reasons, it is of pivotal importance to properly assess flood and droughts risks to identify sustainable solutions.

This course will address all these issues following an iterative cycle consisting of a series of steps with the aim of

minimizing the impact of disasters on communities and enhancing resilience. First, we will start exploring the different methods for assessing the intensity and severity of floods and droughts, i.e. the hazard components of the risk. Second different approaches for floods and droughts risk management will be described, including a game on how to manage multi-risk. Third, we will focus on assessing the impacts of floods and droughts and different sectors, such as water quality and society. Fourth, different adaptation strategies for floods and droughts will be analysed and compared. Finally, the course will address the crucial issue of the dynamics emerging from the interaction between human and physical systems. Although each lecture is associated with one of the topics, most of them are cross-disciplinary and will deal with more than one of these concepts

Additional Information Teaching Methods

The course sessions consist of topical lectures by relevant experts, including two computer practicums for the assessment of water extremes and their impacts, panel discussions, and games on water extremes management. Lectures will be on-campus, except where specified in the Schedule section below; any Zoom links are posted on the Canvas page. Different experts will be invited to discuss the impact of floods and droughts on human health, water quality, and water availability

It is important that you attend all sessions. Although attendance is not formally checked, please make sure you participate, because the interaction in the classroom is a highly valuable part of the course. Do not fall behind with following lectures and reading the suggested materials: this is an intensive course, and you would have a hard time catching up. Additionally, to fully benefit from each lecture and to efficiently participate, it will be best if you have followed the previous ones.

Please communicate to the course coordinator if you are incapacitated to attend any session. At the start of the course, you will team up in groups of three/four students to write a paper. The topic will be water-related issues and adaptation in a city of your choice; you will also present the results during the two final sessions. This course grants 6 credits, implying a study load of 168 hours. The table below breaks down a rough subdivision of the study load into the different activities.

Method of Assessment

The course is assessed through 1) a written exam; 2) a paper assignment; and 3) a peer review of another group’s paper.

1. Written exam (60%): Will last about 2 hours. It will be ‘closed book’, it will contain both open and multiple-choice questions, and it will be based on the readings (excluding the ‘background’ ones) and on the lectures. Also, questions related to insights learned during the computer practicums can be asked. Quantitative questions may be asked, for which you’ll need to bring a calculator. The written exam counts for 60% of the final grade. Regardless of the average final grade, the minimum exam score to pass the course is 5 out of 10.
2. Paper assignment (35%): Each group of students will select a city or region of their choice and investigate a water extreme issue, its risks, and solutions. Each group will then compose and present a paper. Paper and presentation will be evaluated as 35% of the final grade. Regardless of the average final grade, the minimum paper score to pass the course is 5 out of 10.
3. Peer review (5%): You will also individually perform a peer review of the paper of another group, and you will be graded for the quality and insights of your review, representing 5% of the final grade.

The final grade will be calculated given the proportions above and rounded to the half grade. Other small exercises, although not evaluated, are compulsory, these are: The exercise of the computer practicums. These need to be handed in via Canvas. You will receive short, collective feedback on them. Presence and active contribution to all student presentations

Entry Requirements

There are no specific requirements, besides enrollment in the MSc Environment and Resources Management or other MSc for which the course is a possible elective.

Literature

Readings for each lecture are specified on the Canvas page. They consist of book chapters and papers and aim to give you a broad understanding of the water management issues as befits a master’s program. It is very advisable to familiarize with the readers before the lecture and to do any preparatory exercises suggested in Canvas. This helps you to actively engage in discussions during our meetings, to think of the questions you want to ask the lecturer, and also makes it easier to prepare for the final exam. The main notions of the readings can be tested during the exam. Some items are specified as “background reading”, and they are meant as suggestions to explore specific subjects in higher detail. All readings are either freely available online from university computers and from home by using the VU proxy server, or are uploaded to Canvas.

Additional Information Target Audience

MSc students Environment and Resource Management (ERM); MSc Hydrology; Global Environmental Change & Policy (GEC&P); Earth Sciences (ES)

Water Politics and Governance

Course Code	AM_1192
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J. Dell'Angelo
Examiner	dr. J. Dell'Angelo
Teaching Staff	dr. J. Dell'Angelo
Teaching method(s)	Lecture

Course Objective

Water is central for ecosystems, human wellbeing and development. The relationship between society and water resources is critical and complex. Water issues have been historically treated as a technical problem, something to be resolved through hydraulic and engineering solutions. However, the increasing and exacerbating competition over water resources due to expanding global population, climate change and conflicting production activities raises new political and democratic challenges. Understanding water governance implies developing multidisciplinary knowledge about the different problems, approaches, perspectives and controversies related to how societies cooperate and compete over water resources. This course aims at building students' knowledge, critical thinking and analytical skills on the governance of freshwater resources with particular emphasis on the institutional, political and organizational dimensions.

Course objectives:

The main objective of this course is for you to develop an expert understanding of the multidimensional challenges of governance of water resources. Collectively as a class, and individually, you will develop expertise in the water governance world by engaging with the core scientific literature, discussing and tackling real world problems and independent research activities. By the end of the course, you will have gained an understanding of the complexity of water governance: the various models of governance, its structures and institutions at multiple scales, the different values and uses of water, and the main frames and debates from multiple perspectives.

By the end of this course you will be able to:

1. Describe various principles and theories for water governance and □management
2. Apply these to analyze different water policy and governance interventions, by analyzing them from diverse and diverging perspectives
3. Formulate your own analysis and suggestions regarding water-governance interventions on multiple scales: local, national and international
4. Synthesize and communicate your ideas and expertise orally and in written form

Course Content

Tentative program which can be revised, main themes are:

- Water politics and critical perspectives
- Neoliberalism and privatization
- Commons Theory and Community Irrigation Schemes
- Water Law and Water Rights
- NEXUS, Virtual Water, Water & Energy, Water and Mining
- International cooperation and water conflicts
- Global virtual water trade, water grabbing, water war and peace

Additional Information Teaching Methods

This course will be intensive, during three weeks you will be "immersed" in water governance studies. We will spend an important amount of the course in the classroom but you will be primarily required to study the readings to prepare for lectures and discussions. We will have three weekly meetings and each day we will mix: frontal lectures, discussions and group activities.

The structure of each session, generally, consists of lectures at the start, where we will first highlight concepts and key questions related to the topics we are working on. We will then engage in-group discussions associated with the key issues addressed in each session. The group activities will include simulations and team assignments and presentations. Readings will be assigned for each session and it will be essential to be well prepared in order to successfully participate and address key questions that we will tackle in the discussions. Finally, writing exercises, group activities, and debates will be an integral part of the course too.

Participation and preparation: Due to the complexity of the issues addressed and the concentrated length of the course I highly recommend attending every session. Of course, if you will have serious family or health reasons you can count on our help for making up for your absence and catching up, but otherwise it will really not be in your interest to miss classes. You are kindly invited inform the teaching assistant in case you cannot attend one session. The amount of the material that we will go through and the very short time to prepare between the end of classes and the final exam will not allow you to be successful in this course if you procrastinate.

We will have group discussions on key questions related to the readings lead by myself and other teaching assistants. In the discussions and lectures we will call on students directly to address specific question during the group discussion. Referring to the “Socratic method” you might be challenged to defend your informed opinion and perspective on specific subjects. (This is a method that might seem a little intimidating, but is particularly useful to develop the capacity to debate publicly and defend your opinions – in line with the learning goals of the course).

Preparation advice: As MSc students you are very qualified, motivated and brilliant therefore some of this advice might be unnecessary. However, I can’t stress enough the importance to hit the ground and start running from day one, actively participate and not fall behind...keep in mind it’s only 3 weeks of classwork. A critical aspect is that you make an effort to study the assigned readings before class. If you don’t you will see pages adding up quickly and you not be as able to engage productively in discussions and have less information and tools to understand the lectures.

Method of Assessment

- Class participation, public debated, discussion and presentations.
- Water documentary review: This is a team project that includes a. search and selection of water documentaries; b. a written review of one selected documentary; c. a presentation to the class.
- In class final exam: there will be a closed-book 2.15h digital exam prevalently with open ended questions.

Literature

Selected articles and book chapters available in Canvas, organized around the above topics. There is no textbook for this class. For the most part, we will use scientific peer reviewed articles and, in certain cases, policy reports or other kind of documents. As we will be using scientific articles I will provide instructions on what I expect you should be focusing.

There will also be an additional number of suggested readings for the people interested in learning more about specific topics. Also, please note that the reading list available here in some rare cases might change during the course but you will all be clearly notified. You will find the final readings list every week and all the required readings in weekly folders on Canvas.

Additional Information Target Audience

Master level students with an interest in understanding the role of governance in the emergence of water issues, and their potential resolution

Additional Information

The course is coordinated and taught by Dr. Jampel Dell'Angelo (Associate Professor of Water Governance and Politics) with invited guest speakers. Language of teaching is English.

Custom Course Registration

Through the regular VU channels

Recommended background knowledge

no specific background required, just an eager motivation to engage, debate, learn and READ (a lot)